More Images, More Problems A Controlled Analysis of VLM Failure Modes (2601.07812v1)

Abstract

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> Large Vision Language Models (LVLMs) have demonstrated remarkable capabilities, yet their proficiency in understanding and reasoning over multiple images remains largely unexplored. While existing benchmarks have initiated the evaluation of multi-image models, a comprehensive analysis of their core weaknesses and their causes is still lacking.

Why MIMIC and corresponding solution was proposed.

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> : LVLMs often fail to aggregate information across images and struggle to track or attend to multiple concepts simultaneously.

Main failure mode is the lack of ability to to make sense of multiple concepts occuring in multiple images simultaneously.

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> On the data side, we present a procedural data-generation strategy that composes single-image annotations into rich, targeted multi-image training examples. On the optimization side, we analyze layer-wise attention patterns and derive an attention-masking scheme tailored for multi-image inputs

Two solutions proposed:

1. Converting single-image single-annotation to into dense, diverse single-image multi-annotation dataset.
2. Attention masking mechanism tailored for multi-image input.

Conclusion

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>
> > Our analysis reveals that current SOTA models fundamentally exhibit “single-
>
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> >
> > image behavior,” struggling to aggregate information across inputs or track multiple concepts in the presence of visual distractors
>
> Main finding. Also LVLMs are susceptible to distractors, a similar finding has been done in previous works.

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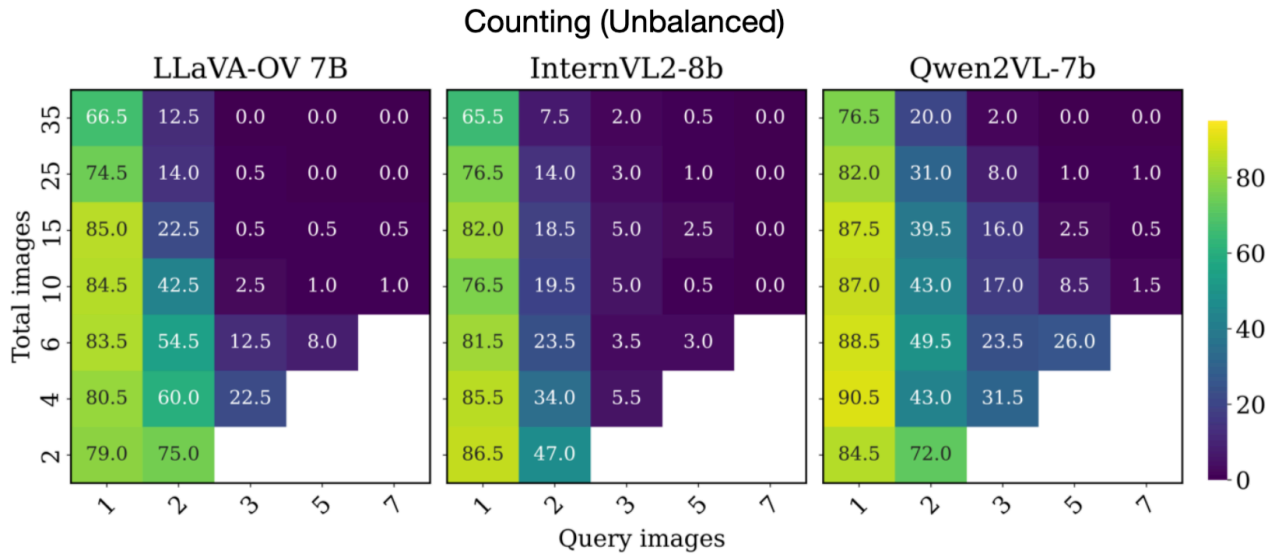
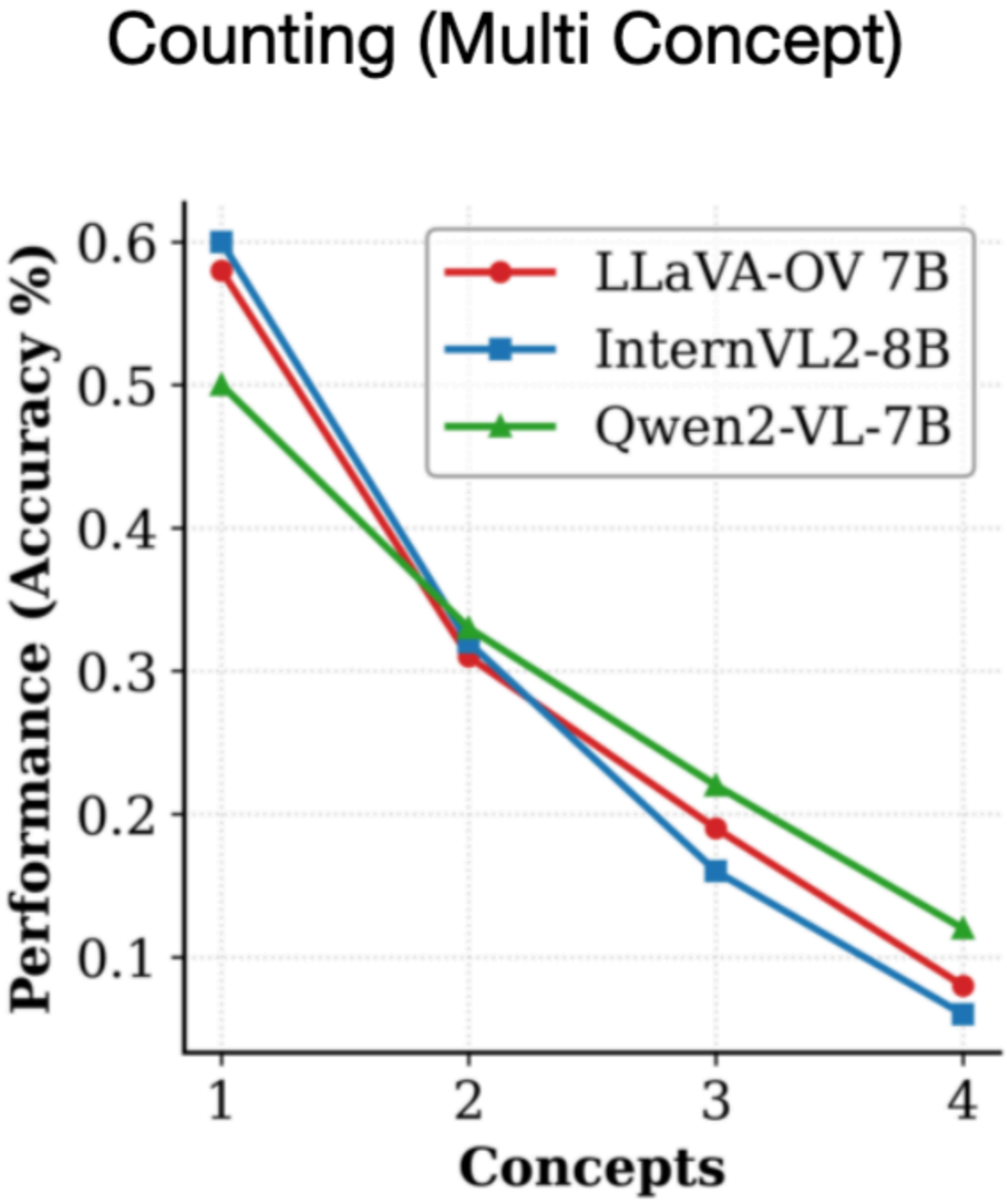
>
> > o address this, we introduced a data-centric synthetic fine-tuning strategy and an optimization-centric attention-masking mechanism
>
> 2 solutions proposed by authors. New SOTA

Tables and figures



Performance on counting task degrades as the number of query images increase.



Multi-concept tracking performance is even worse.

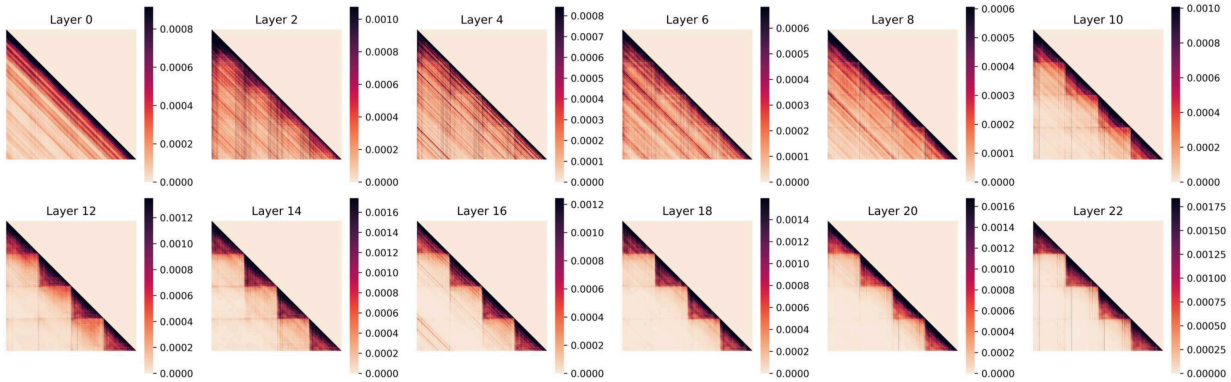
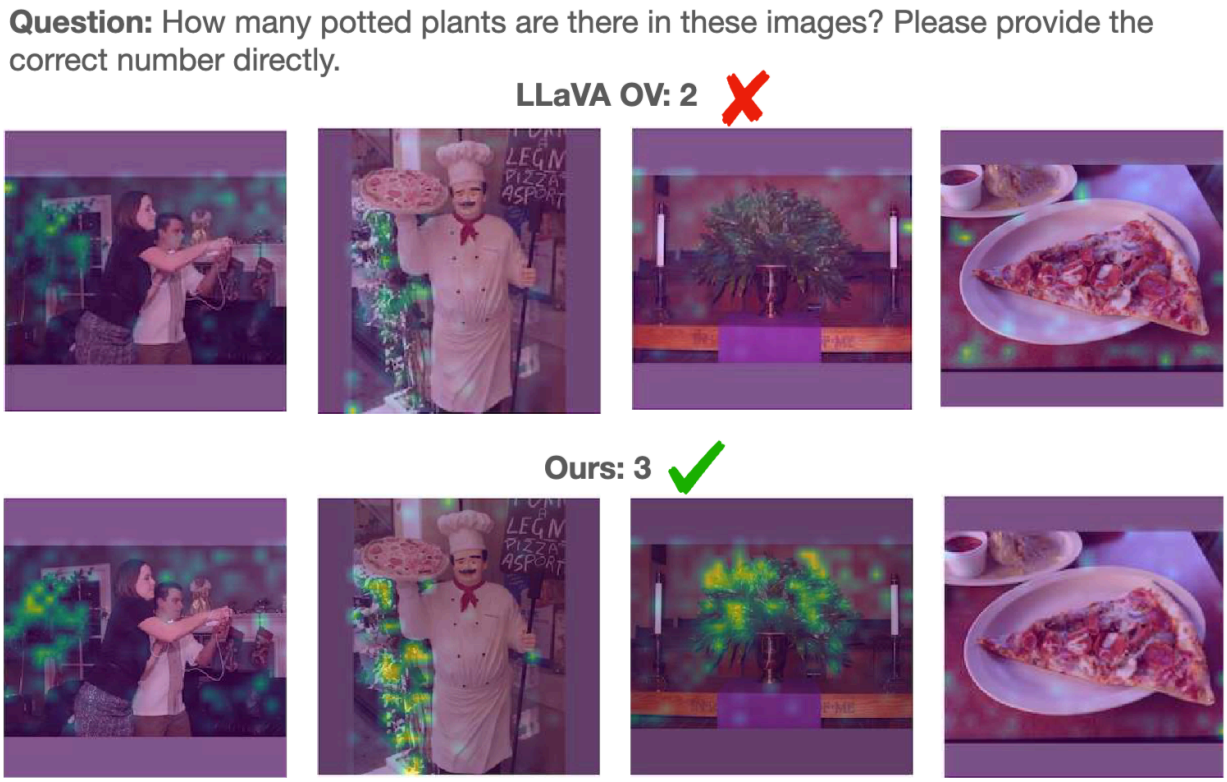


Figure 4: **Inter-image and intra-image token attention across layers.** The attention patterns transitions from cross-image to intra-image interactions as we advance in depth.

A clear pattern in attention is observed. In the earlier layers, the attention pattern is distributed equally. Thus we see the diagonal and parallel lines. Then, as we move to the later layers, we can clearly see 4 triangles getting formed, indicating that the model is now attending to tokens within the image itself.



After applying the solution, the model is now able to focus on the required key or object.

Introduction

LVLMs face significant challenges. Identification and addressing these challenges follows two pathways:

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> he development of comprehensive evaluation benchmarks (Goyal et al., 2017; Mathew et al., 2021; Kembhavi et al., 2016; Masry et al., 2022; Li et al., 2024b; Fu et al., 2024a) and the study of the models' inner workings

Two pathways for identifying and addressing challenges faced by the models.

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> In this work, we address this gap by conducting a systematic study of LVLMs in multi-image contexts.

Most work focuses on single-image setting. Hence authors propose this benchmark which uses multi-image setting.

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> MIMIC procedurally generates multi-image sequences by leveraging per-image annotations that give fine-grained control over information spread, distractor presence, object-instance

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>
> distributions, sequence length, and query complexity, while providing unambiguous ground-truth answers for robust, decorrelated analysis of the model's strengths and weaknesses

Benchmark is constructed using MS COCO. It is annotated well but limited to 80 classes.

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> Using both quantitative and qualitative assessments, our study reveals that current state-of-the-art LVLMs struggle to effectively aggregate information across multiple images, are unable to track/attend to multiple concepts simultaneously, while being susceptible to distractors

Finding, same as stated above.

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> We attribute these shortcomings to a combination of factors, including limitations in multi-image sequence modeling, training data biases, poor inter-image communication induced by the causal attention and the inherent complexity of multi-image reasoning tasks

Causes of the problems mentioned above.

Method

The tasks are:

1. Counting: Counting the instances of objects
2. Listing: List all sub-categories for a given parent category.
3. Odd one: Finding the different image from among the similar ones
4. Common one: Determining a class common across the given images.

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> o minimize the impact of potential class imbalance, we first select a pool of classes and then sample from this pool, ensuring that each class is chosen with an equal probability. This ensures that the distributions of classes and instances remain consistent across settings.

Authors perform uniform sampling from the classes.

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> We refer to this as the information spread

Counting task contains information spread, where the queried classes can be concentrated on one image or across multiple images.

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> For instance, increasing the number of object classes k tests the model's multi-concept tracking ability, while varying the information spread evaluates its capacity to aggregate information across images.

Authors give an example of what insights can each dimension can give.

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> his task evaluates the model's ability to exhaustively extract information in a dense manner from multiple images. As a byproduct, it also measures the model's visual perception ability to recognize and categorize multiple objects, as well as its capacity to aggregate this information into a coherent list

Listing tasks should extract sub-classes, also the ability to recognize classes when occluded or spread across images is also tested. This involves multiple classes.

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> he two tasks are designed to assess the model's ability to identify shared or unique elements across multiple images. Importantly, while previous tasks focus on aggregating information, these tasks require comparative reasoning across images, hence the model must first implicitly identify all objects before performing cross-image analysis.

For common and odd-one, cross-image analysis becomes necessary before answering the question.

Findings

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> The control experiment confirms that gains are due to sequence length reduction rather than information reduction. This suggests that the models primarily struggle with long sequence understanding rather than with processing multiple distinct images.

Preserving the number of images but downsampling them preserves information but reduces resolution. This leads to smaller number of tokens. In this setting the performance incerases, indicating performance drop due to increased sequence length.

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> Finding 2: Current LVLMs primarily behave as single-image models: performance peaks when the vision-token sequence length matches that produced by one or two images.

This is true for LLAVA-OV models.

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> vary the information spread in the counting task, which defines how object instances are distributed across images. In Fig. 1 (left and middle) we show the results of increasing the number of images containing the object instance from 1 to 7. We observe a sharp accuracy drop that approaches 0 even when very few distractors are present

This indicates that the models cannot track and aggregate the queried object across multiple images. When queried entities are spread across multiple images, all models fail in this scenario.

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> Finding 4: Models are sensitive to visual distractors, especially if information is spread out

Models are susceptible to distractors and spread of information.

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> Finding 5: LVLMs demonstrate limited capacity for multi-concept tracking, reducing their

Models fail on multi-concept tracking.

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> To probe how visual information is propagated and integrated across images at the token level, we analyze attention patterns among vision tokens in multi-image inputs. Concretely, we compute the normalized attention scores from each vision token to all other vision tokens in the input sequence subject to an autoregressive attention masking on a subset of 50 samples

Experiment for attention pattern.

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> n earlier layers, there is a significant amount of inter-image attention, indicating that the model is attempting to integrate information across images. However, as we progress to deeper layers, the attention becomes predominantly intra-image. This inflection point occurs somewhere around the middle of the network. This shift may contribute to the observed difficulty in aggregating information across multiple images. Conceptually, the build-up of representations appears to proceed from broad semantic correlations across images to finer-grained, instance-level integration

Conclusions from attention pattern analysis.

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> This has a series of consequences: (1) the early inter-image attention may introduce noise or distractions that hinder the model's ability to focus on relevant information in later layers; hence, early mistakes in cross-image interaction are harder to correct; (2) the cross-image interaction under a causal attention mechanism may lead to error propagation, where tokens belonging to later images cumulate higher amount of noise with incorrect information from earlier images; this may reduce the vision perception capability of the model for later images and explain some of the performance degradation as the number of images increases; (3) the architecture and training objectives may not sufficiently encourage cross-image integration, leading to a default behavior of treating images independently; (4) the observed attention patterns may reflect inherent biases in the training data, where the multi-image tasks don't require deep cross-image reasoning, leading the model to learn shortcuts that prioritize single-image understanding

More conclusions from attention pattern analysis. This has potential for ideas.

Solutions by authors

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> Multi-Image Finetuning:

This contains data sampled from OpenImages and has tasks similar to MIMIC for diversity.

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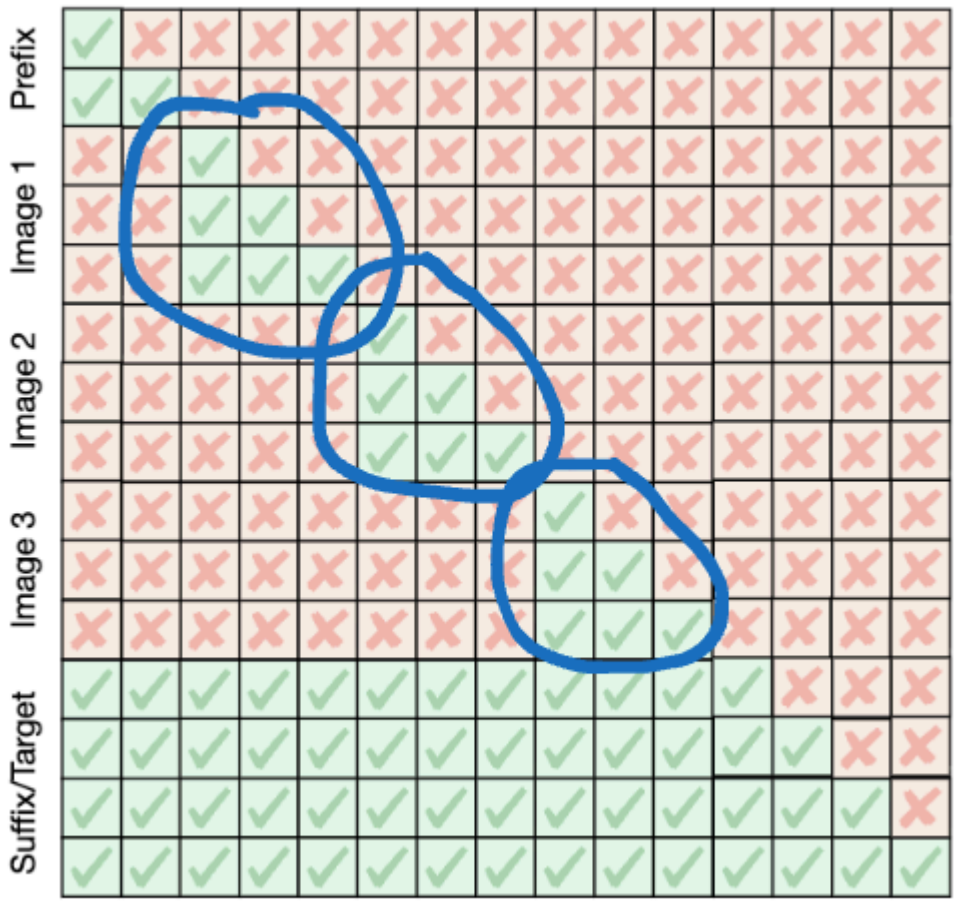
> we apply layer-wise attention masking during fine-tuning, restricting vision tokens to attend only to tokens from the same image in selected layers, while leaving text-token attention unchanged. This design offers two key benefits. First, it reduces unnecessary cross-image interactions, leading to a more efficient model with lower computational cost (see table 7 and fig. 7 of appendix). Second, it encourages cleaner imagelocal representations in deeper layers, which empirically improves performance across several benchmarks. For this setting, we employ LoRA-based fine-tuning to further improve parameter efficiency.

Layer wise attention mask applied such that the visual tokens attend to other visual tokens of the same image. This reduces cross image interaction. This also improves local representations in deeper layers.

Question: We observed that attention becomes increasingly inter-image in deeper layers. With the current solution, we can restrict the attention of visual tokens to visual tokens within the same image, but what will be its effect on *spread of information, information aggregation* as these tasks require information from multiple images?
Answer: Maybe it reduces useless information aggregation.



If we observe the solution, we can see these triangles being formed. This means that visual tokens can attend to the tokens within same image and the prefix. Not to any other image and since it is causal mechanism, not attend to any tokens after.